CS341 · COMPUTER GRAPHICS

Mock Midterm Practice

20 marks · the same distribution as the exam

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01 Questions

Q1 TRUE / FALSE · 5 PTS

1. **In a raster scan system, picture information is stored in the frame buffer.** ( )
2. **A plasma panel is a nonemissive display.** ( )
3. **Persistence is the time for the emitted light to decay to one-half of its original intensity.** ( )
4. **The difference between two points is a vector.** ( )
5. **In DDA, when |m| > 1 we step x by one unit and compute y.** ( )

Q2 MCQ · 5 PTS

1. **Which of the following is a hard-copy device?**
   a) Trackball b) Ink-jet printer c) Light pen d) Data glove
2. **For crisp images, the dot pitch should be less than:**
   a) 0.13 mm b) 0.31 mm c) 3.1 mm d) 1.3 mm
3. **Two vectors are perpendicular when their dot product is:**
   a) 1 b) −1 c) 0 d) Their length

4. The frequency at which a picture is redrawn on the screen is:

a) Resolution b) Persistence c) Refresh rate d) Dot pitch

5. In OpenGL, the function that sets the 2D world coordinate reference frame is:

a) glVertex2i b) gluOrtho2D c) glutInitWindowSize d) glClearColor

Q3 SHORT ANSWER · 4 PTS

1. List four input devices.

2. Why do lines on a raster display look stair-stepped (jaggies)?

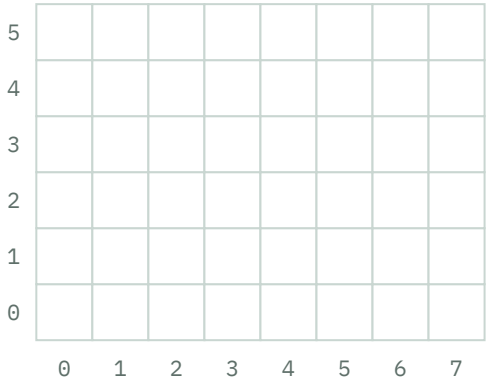
3. Compare random scan and raster scan (drawing, storage, resolution).

4. Compare active-matrix and passive-matrix displays.

Q4 DDA · 5 PTS

Based on the DDA algorithm, calculate the points between the starting point (1,1) and the ending point (6,4). Fill the table, then draw the line using these points.

k	x	y	round(x)	round(y)
0				
1				
2				
3				
4				
5				



Q5 ASPECT RATIO · 1 PT

A screen is 8 inches wide and 6 inches high, with a resolution of 1024 × 768 points. Find the aspect ratio.